COMPACTNESS AND INERTIA, WITH SOLUTIONS

SANTIAGO ARANGO-PIÑEROS

Let E be an elliptic curve over a local field K. The rational points of E with nonsingular reduction form a subgroup $E_0(K)$ of finite index in $E(K)$ [Sil09, Theorem VII.6.1 and Corollary VII.6.2], and E has good reduction if and only if the inertia group acts trivially on its torsion, by the criterion of Néron–Ogg–Shafarevich [Sil09, Theorem VII.7.1]. Problem 1.8 proves the finiteness of $E(K)/E_0(K)$ by a compactness argument. Problem 2.12 studies how the inertia group acts on the torsion of a curve with potential good reduction. Each problem is preceded by the definitions and results it uses.

Standing notation.

- A finite extension K of $\mathbb{Q}_p$, with normalized valuation v, ring of integers $\mathcal{O}_K$, uniformizer π and residue field $k = \mathcal{O}_K/\pi\mathcal{O}_K$, a finite field of characteristic p. (Silverman's Chapter VII assumes that K and k are perfect, and here both are.)
- A fixed algebraic closure $\overline{K}$ of K.
- A bar denotes reduction modulo π.
- An elliptic curve E over K, given by a minimal Weierstrass equation $F(X, Y, Z) = 0$, with $F \in \mathcal{O}_K[X, Y, Z]$ homogeneous of degree 3. It embeds E in $\mathbb{P}^2$ as $V_+(F)$.
- The primitive coordinates $[X : Y : Z]$ of $P \in \mathbb{P}^2(K)$: coordinates in $\mathcal{O}_K$, not all in $\pi\mathcal{O}_K$. Every P has them, and they are unique up to a factor in $\mathcal{O}_K^\times$.
- The reduction $\bar{P} := [\bar{X} : \bar{Y} : \bar{Z}] \in \mathbb{P}^2(k)$ of $P \in \mathbb{P}^2(K)$, computed from primitive coordinates, hence well defined.
- The reduced curve E_π, defined by $\bar{F} = 0$. If $P \in E(K)$, then $\bar{P} \in E_\pi(k)$.

How to use this. Allow about 25 minutes for each problem. The `Magma` block in Problem 2.12 runs in seconds in the free online calculator. Everything else is by hand.

1. Finiteness by compactness

This section is [Sil09, Exercise 7.6]. The argument uses that k is finite, so it does not apply, for instance, to the completion of the maximal unramified extension of $\mathbb{Q}_p$: a field complete with respect to a discrete valuation, with residue field $\overline{\mathbb{F}}_p$ (item (6)). Yet this is the case that the criterion of Néron–Ogg–Shafarevich needs: Silverman's proof of it uses the finiteness of $E(K^{\mathrm{unr}})/E_0(K^{\mathrm{unr}})$, where K^{unr} is the maximal unramified extension of K, whose residue field is an algebraic closure of k [Sil09, proof of Theorem VII.7.1].

Definition 1.1 (The topology of K)**.** We give K the topology defined by v: the sets $a + \pi^n\mathcal{O}_K$, for $a \in K$ and $n \in \mathbb{Z}$, form a basis, and each of them is both open and closed. With it, K is a Hausdorff topological field.

Definition 1.2 (The subgroup $E_0(K)$)**.** Let E_π^{ns} be E_π minus its singular point, if it has one. Then

$$E_0(K) := \{P \in E(K) : \bar{P} \in E_\pi^{\mathrm{ns}}(k)\}.$$

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Definition 1.3 (Regular functions [Poo06, Definition 1.8.3])**.** A function $\psi \in K(E)$ is regular at $Q \in E(\overline{K})$ if $\psi = g/h$ with $g, h \in K[X, Y, Z]$ homogeneous of the same degree and $h(Q) \neq 0$. Then $\psi(Q) \coloneqq g(Q)/h(Q)$, an element of K if $Q \in E(K)$. The functions regular at Q form a discrete valuation ring with fraction field $K(E)$.

Proposition 1.4 ($E_0(K)$ is a subgroup [Sil09, Proposition VII.2.1])**.** *The set $E_0(K)$ is a subgroup of $E(K)$.*

Proposition 1.5 (Compactness [Poo17, §1.1.2])**.** *The ring $\mathcal{O}_K$ is compact: it is complete, and for every n it is covered by the $(\#k)^n$ translates of $\pi^n\mathcal{O}_K$. Equivalently, K is locally compact.*

Proposition 1.6 (Morphisms $E \to E$ over K)**.**
(1) The morphisms $\alpha\colon E \to E$ defined over K are exactly the points $\alpha = [\psi_0 : \psi_1 : \psi_2] \in E(K(E))$, that is, $\psi_0, \psi_1, \psi_2 \in K(E)$, not all zero, with $F(\psi_0, \psi_1, \psi_2) = 0$ [Poo17, Proposition 3.6.5(b)].
(2) If $Q \in E(\overline{K})$ and ψ_0, ψ_1, ψ_2 are regular at Q and not all zero there, then $\alpha(Q) = [\psi_0(Q) : \psi_1(Q) : \psi_2(Q)]$ [Sil09, §I.3].
(3) Replacing each ψ_i by $\phi\psi_i$, for $\phi \in K(E)^\times$, gives the same point of $E(K(E))$, and for every $Q \in E(\overline{K})$ some ϕ makes $\phi\psi_0, \phi\psi_1, \phi\psi_2$ regular at Q and not all zero there: a power of a uniformizer of the discrete valuation ring of Definition 1.3 will do [Sil09, proof of Proposition II.2.1].

Proposition 1.7 (Translations [Sil09, Theorem III.3.6 and Example III.4.7])**.** *Let $P \in E(K)$. The translation $\tau_P\colon E \to E$, $Q \mapsto Q + P$, is an isomorphism defined over K, with inverse τ_{-P}.*

Problem 1.8 ($E(K)/E_0(K)$ is finite)**.**
Let $q\colon K^3 - \{0\} \to \mathbb{P}^2(K)$ be the quotient map. Give $K^3 - \{0\}$ the product topology and $\mathbb{P}^2(K)$ the quotient topology: $U \subseteq \mathbb{P}^2(K)$ is open if and only if $q^{-1}(U)$ is open. Let

$$S \coloneqq \{\mathbf{x} = (x_0, x_1, x_2) \in \mathcal{O}_K^3 : x_i \in \mathcal{O}_K^\times \text{ for some } i\}$$

be the set of primitive coordinate vectors.

(1) Show that S is compact and open in $K^3 - \{0\}$, and that $q(S) = \mathbb{P}^2(K)$. Deduce that $\mathbb{P}^2(K)$ is compact. Show that $U \subseteq \mathbb{P}^2(K)$ is open if and only if $q^{-1}(U) \cap S$ is open in S.

Solution. $S = \mathcal{O}_K^3 - (\pi\mathcal{O}_K)^3$. By Proposition 1.5 and Tychonoff's theorem, $\mathcal{O}_K^3$ is compact, and $(\pi\mathcal{O}_K)^3$ is open, so S is closed in a compact space, hence compact. It is also open: S is the union over i of the sets $\mathcal{O}_K \times \mathcal{O}_K \times \mathcal{O}_K$ with the i-th factor replaced by $\mathcal{O}_K^\times$, and $\mathcal{O}_K^\times$ is open, being the union of the balls $u + \pi\mathcal{O}_K$ with u a unit. If $\mathbf{x} \in K^3 - \{0\}$ and $m = \min_i v(x_i)$, then $\pi^{-m}\mathbf{x} \in S$ and $q(\pi^{-m}\mathbf{x}) = q(\mathbf{x})$, so $q(S) = \mathbb{P}^2(K)$. As q is continuous, $\mathbb{P}^2(K) = q(S)$ is compact.
If U is open, then $q^{-1}(U) \cap S$ is open in S. Conversely, suppose that $V \coloneqq q^{-1}(U) \cap S$ is open in S, hence in $K^3 - \{0\}$, since S is open. Every $\mathbf{x} \in K^3 - \{0\}$ is $c\,\mathbf{s}$ with $c \in K^\times$ and $\mathbf{s} \in S$, and $q^{-1}(U)$ is stable under $K^\times$, so

$$q^{-1}(U) = \bigcup_{c \in K^\times} cV.$$

Multiplication by c is a homeomorphism of $K^3 - \{0\}$, so this is a union of open sets, and U is open.

(2) Show that $E(K) = V_+(F)(K)$ is closed in $\mathbb{P}^2(K)$, and deduce that it is compact.

Solution. The set $q^{-1}(E(K)) = \{\mathbf{x} \in K^3 - \{0\} : F(\mathbf{x}) = 0\}$ is closed, because F is a continuous function $K^3 \to K$ (K is a topological field) and $\{0\}$ is closed in the Hausdorff space K. So $E(K)$ is closed in the compact space $\mathbb{P}^2(K)$, hence compact.

(3) Let $g, h \in K[X, Y, Z]$ be homogeneous of the same degree. Show that $W := \{Q \in \mathbb{P}^2(K) : h(Q) \neq 0\}$ is open, and that $W \to K$, $Q \mapsto g(Q)/h(Q)$, is continuous. Deduce from Propositions 1.6 and 1.7 that every translation $\tau_P \colon E(K) \to E(K)$ is a homeomorphism.

Solution. As in (2), the set where h vanishes is closed, so W is open. The function $G(\mathbf{x}) := g(\mathbf{x})/h(\mathbf{x})$ is continuous on the open set $q^{-1}(W)$, and $G(c\mathbf{x}) = G(\mathbf{x})$ for $c \in K^\times$, since g and h have the same degree. So $G = \theta \circ q$ for a function $\theta \colon W \to K$. If $V \subseteq K$ is open, then $q^{-1}(\theta^{-1}(V)) = G^{-1}(V)$ is open in $q^{-1}(W)$, which is open in $K^3 - \{0\}$. So $\theta^{-1}(V)$ is open, and θ is continuous.
Let $P, Q \in E(K)$. By Proposition 1.7, τ_P is a morphism defined over K. Write $\tau_P = [\psi_0 : \psi_1 : \psi_2]$ as in Proposition 1.6, with the ψ_i regular at Q and $\psi_j(Q) \neq 0$ for some j (part (3)). Write $\psi_i = g_i/h_i$ with $h_i(Q) \neq 0$ (Definition 1.3). On the open neighborhood $U := E(K) \cap \{h_0 h_1 h_2 \neq 0\}$ of Q in $E(K)$, each ψ_i is regular and, by the first part, continuous. So $U' := \{Q' \in U : \psi_j(Q') \neq 0\}$ is an open neighborhood of Q. On U', by part (2), τ_P is the composition of the continuous maps

$$U' \to K^3 - \{0\},\ Q' \mapsto \big(\psi_0(Q'), \psi_1(Q'), \psi_2(Q')\big), \qquad \text{and} \qquad q \colon K^3 - \{0\} \to \mathbb{P}^2(K).$$

So τ_P is continuous on each set of an open cover of $E(K)$, hence continuous. Its inverse τ_{-P} is continuous for the same reason, so τ_P is a homeomorphism.

(4) Show that the reduction map $\mathbb{P}^2(K) \to \mathbb{P}^2(k)$, $P \mapsto \bar{P}$, is locally constant. Deduce that $E_0(K)$ is open in $E(K)$, and also closed.

Solution. For $\mathbf{x} \in S$, the open neighborhood $\mathbf{x} + (\pi\mathcal{O}_K)^3$ of $\mathbf{x}$ lies in S, and every $\mathbf{x}'$ in it has $\bar{\mathbf{x}}' = \bar{\mathbf{x}}$. So $\mathbf{x} \mapsto \overline{q(\mathbf{x})}$ is locally constant on S. Hence for every subset $T \subseteq \mathbb{P}^2(k)$, the set $q^{-1}(\{P : \bar{P} \in T\}) \cap S$ is a union of such neighborhoods, so open in S, and $\{P : \bar{P} \in T\}$ is open by (1). Its complement is the set for $\mathbb{P}^2(k) - T$, so it is also closed. With $T = E_\pi^{\mathrm{ns}}(k)$, $E_0(K)$ is the intersection of $E(K)$ with an open and closed subset of $\mathbb{P}^2(K)$.

(5) Prove that $E(K)/E_0(K)$ is finite.

Solution. By Proposition 1.4, the cosets $P + E_0(K) = \tau_P(E_0(K))$ partition $E(K)$, and each is open by (3) and (4). By (2), $E(K)$ is compact, so finitely many cosets cover it. Since the cosets are disjoint and nonempty, each one must appear in this finite subcover, so there are finitely many.

(6) Let K' be a field complete with respect to a discrete valuation whose residue field k' is infinite, for instance algebraically closed. Show that its ring of integers is not compact. Which step of the argument above fails for an elliptic curve over K'?

Solution. **The first step, (1): $\mathbb{P}^2(K')$ is not compact.** Choose lifts $a_\alpha \in \mathcal{O}_{K'}$ of the elements $\alpha \in k'$. The balls $a_\alpha + \pi\mathcal{O}_{K'}$ are open, pairwise disjoint and cover $\mathcal{O}_{K'}$, and there are infinitely many of them. Since none can be removed, $\mathcal{O}_{K'}$ is not compact. So the first step fails: $\mathcal{O}_{K'}^3$, S and $\mathbb{P}^2(K')$ are not compact, and for an elliptic curve E' over K', the cover of $E'(K')$ by the cosets of $E_0'(K')$ need have no finite subcover.

Remark. The finiteness of $E'(K')/E_0'(K')$ is still true, but it needs the theory of Néron models [Sil09, Corollary VII.6.2].
Compare [Poo17, §2.6.2 and Proposition 2.6.1], which define the same kind of topology on $X(K)$ for every variety X over K, and show that $X(K)$ is compact when X is proper and K is a local field.

2. Inertia acting on torsion

This section is [Sil09, Exercise 7.9], with the hypothesis $m \geqslant 3$ added to its first two parts (item (5) explains why) and a `Magma` check. We use the notation of [Sil09, §VII.4].

Definition 2.1 (Reduction types)**.** The curve E has good reduction if E_π is nonsingular, that is, if $v(\Delta) = 0$, and bad reduction otherwise. It has potential good reduction if it has good reduction over some finite extension of K.

Definition 2.2 (Inertia)**.** Let $\mathrm{Gal}_K := \mathrm{Gal}(\overline{K}/K)$, $K^{\mathrm{unr}} \subseteq \overline{K}$ the maximal unramified extension of K, and $I_v := \mathrm{Gal}(\overline{K}/K^{\mathrm{unr}})$ the inertia group. A set on which Gal_K acts is unramified if I_v acts trivially on it.

Definition 2.3 (Ramification of finite extensions)**.** For an extension L of K inside $\overline{K}$, let $\mathrm{Gal}_L := \mathrm{Gal}(\overline{K}/L)$. If L/K is finite and Galois, the inertia group of L/K is the image of I_v in $\mathrm{Gal}(L/K)$, isomorphic to $I_v/(I_v \cap \mathrm{Gal}_L)$. The extension L/K is unramified if its inertia group is trivial, and tamely ramified if its order is prime to p.

Definition 2.4 (Torsion fields and Tate modules [Sil09, §III.7])**.** For $m \geqslant 1$ with $p \nmid m$, $K(E[m])$ is the field generated over K by the coordinates of the points of $E[m](\overline{K})$, so $\mathrm{Gal}_{K(E[m])}$ is the kernel of the action of Gal_K on $E[m](\overline{K})$. For a prime $\ell \neq p$, the Tate module $T_\ell(E)$ is the inverse limit of the $E[\ell^n](\overline{K})$ along the surjections $[\ell]$, and $\rho_\ell\colon \mathrm{Gal}_K \to \mathrm{Aut}\, T_\ell(E)$ is the action of Gal_K on it.

In the following results, $\ell \neq p$ is a prime and $m \geqslant 1$ is an integer with $p \nmid m$.

Proposition 2.5 (Minimal equations [Sil09, Remark VII.1.1 and §III.1])**.** *A Weierstrass equation with coefficients in $\mathcal{O}_K$ and $v(\Delta) < 12$ is minimal. For $y^2 = x^3 + Ax + B$, $\Delta = -16(4A^3 + 27B^2)$.*

Corollary 2.6 (Torsion [Sil09, Corollary III.6.4(b)])**.** *The group $E[m](\overline{K})$ is isomorphic to $(\mathbb{Z}/m\mathbb{Z})^2$.*

Proposition 2.7 (The Tate module [Sil09, Proposition III.7.1(a)])**.** *The Tate module $T_\ell(E)$ is isomorphic to $\mathbb{Z}_\ell^2$. Hence each projection $T_\ell(E) \to E[\ell^n](\overline{K})$ is a surjective homomorphism of Gal_K-modules, and its kernel is $\ell^n T_\ell(E)$, since both have index ℓ^{2n}.*

Proposition 2.8 (Good reduction [Sil09, Proposition VII.4.1])**.** *If E has good reduction, then $E[m](\overline{K})$ and $T_\ell(E)$ are unramified.*

Theorem 2.9 (Néron–Ogg–Shafarevich [Sil09, Theorem VII.7.1])**.** *If $T_\ell(E)$ is unramified, then E has good reduction.*

Corollary 2.10 (Potential good reduction [Sil09, Corollary VII.7.3])**.** *The curve E has potential good reduction if and only if $\rho_\ell(I_v)$ is finite.*

Proposition 2.11 (Inertia groups of finite extensions [Sil09, proof of Corollary VII.7.3])**.** *If $H \subseteq I_v$ is a closed normal subgroup of finite index, there is a finite extension K' of K inside $\overline{K}$ whose inertia group $\mathrm{Gal}(\overline{K}/K'K^{\mathrm{unr}}) = I_v \cap \mathrm{Gal}_{K'}$ is H.*

Problem 2.12 (The inertia group of $K(E[m])/K$)**.**
Suppose that E has potential good reduction, and let J be the subgroup of the elements of I_v that act trivially on $E[m](\overline{K})$ for every $m \geqslant 1$ with $p \nmid m$.

(1) Let ℓ be a prime and $r \geqslant 1$ with $\ell^r \geqslant 3$, and let $g \in \mathrm{GL}_2(\mathbb{Z}_\ell)$ be an element of finite order with $g \equiv I \pmod{\ell^r}$. Show that $g = I$.

Solution. Suppose that $g \neq I$. Some power of g has prime order q, and it is still congruent to I modulo ℓ^r, so we may assume that g has order q. Write $g = I + \ell^s h$ with $h \in M_2(\mathbb{Z}_\ell)$ not

divisible by ℓ. Then $s \geqslant r$, so $\ell^s \geqslant 3$. Expanding,

$$0 = g^q - I = q\ell^s h + \sum_{i=2}^{q} \binom{q}{i} \ell^{is} h^i.$$

If $q \neq \ell$, every term of the sum is divisible by ℓ^{2s}, so $q\ell^s h \equiv 0 \pmod{\ell^{s+1}}$ and, q being a unit, $h \equiv 0 \pmod{\ell}$: a contradiction. If $q = \ell$, then ℓ divides $\binom{\ell}{i}$ for $2 \leqslant i < \ell$, so those terms are divisible by ℓ^{2s+1}, hence by ℓ^{s+2}. The last term is divisible by $\ell^{\ell s}$, and $\ell s \geqslant s+2$ because $s(\ell-1) \geqslant 2$: this is where $\ell^s \geqslant 3$ is needed. So $\ell^{s+1} h \equiv 0 \pmod{\ell^{s+2}}$, and again $h \equiv 0 \pmod{\ell}$.

Remark. For $\ell^r = 2$ the statement is false: $g = -I$ has order 2 and $-I \equiv I \pmod 2$.

(2) Show that $J = I_v \cap \ker \rho_\ell$ for every prime $\ell \neq p$, and that J has finite index in I_v.

Solution. Let $J_\ell := I_v \cap \ker \rho_\ell$. An element of J acts trivially on every $E[\ell^n](\overline{K})$, hence on their inverse limit $T_\ell(E)$, so $J \subseteq J_\ell$. Conversely, $\rho_\ell(I_v)$ is finite by Corollary 2.10, so J_ℓ, the kernel of $I_v \to \rho_\ell(I_v)$, has finite index. It is normal, and closed, being the intersection of the kernels of the actions on the finite sets $E[\ell^n](\overline{K})$. By Proposition 2.11, J_ℓ is the inertia group of a finite extension K' of K, again a finite extension of $\mathbb{Q}_p$. Over K', $T_\ell(E)$ is unramified, so E has good reduction over K' (Theorem 2.9). Then $E[m](\overline{K})$ is unramified over K' for every m with $p \nmid m$ (Proposition 2.8), that is, J_ℓ acts trivially on it. So $J_\ell \subseteq J$.

(3) Let $m \geqslant 3$ with $p \nmid m$. Show that $J = I_v \cap \mathrm{Gal}_{K(E[m])}$, and deduce that the inertia group of $K(E[m])/K$ is isomorphic to I_v/J, independently of m.

Solution. By definition, $J \subseteq I_v \cap \mathrm{Gal}_{K(E[m])}$. Conversely, since $m \geqslant 3$, it has a divisor $\ell^r \geqslant 3$ with ℓ prime: an odd prime factor of m, or 4 if m is a power of 2. Then $\ell \neq p$, as $p \nmid m$, and $E[\ell^r] \subseteq E[m]$. Let $\sigma \in I_v \cap \mathrm{Gal}_{K(E[m])}$. Then σ acts trivially on $E[\ell^r](\overline{K}) \cong T_\ell(E)/\ell^r T_\ell(E)$ (Proposition 2.7), so the matrix of $\rho_\ell(\sigma)$ in a $\mathbb{Z}_\ell$-basis of $T_\ell(E)$ is congruent to I modulo ℓ^r. It has finite order, since $\rho_\ell(I_v)$ is finite (Corollary 2.10), so $\rho_\ell(\sigma) = I$ by (1), and $\sigma \in J$ by (2). So $J = I_v \cap \mathrm{Gal}_{K(E[m])}$, and the inertia group of $K(E[m])/K$ is isomorphic to $I_v/(I_v \cap \mathrm{Gal}_{K(E[m])}) = I_v/J$.

(4) Let $m \geqslant 3$ with $p \nmid m$. Show that $K(E[m])/K$ is unramified if and only if E has good reduction. Which direction holds for every $m \geqslant 1$ with $p \nmid m$, and for every elliptic curve over K?

Solution. **Good reduction implies unramified, for all such m and E.** If E has good reduction, then $E[m](\overline{K})$ is unramified by Proposition 2.8, so $K(E[m])/K$ is unramified. This holds for every m with $p \nmid m$ and every elliptic curve over K. Conversely, suppose that $m \geqslant 3$ and that $K(E[m])/K$ is unramified. Then I_v/J is trivial by (3), so $I_v = J = I_v \cap \ker \rho_\ell$ for a prime $\ell \neq p$ by (2). So $T_\ell(E)$ is unramified, and E has good reduction by Theorem 2.9.

(5) Let $p \geqslant 3$ and $E\colon y^2 = x^3 - \pi^2 x$.

(a) Show that E has bad reduction, potential good reduction, and $K(E[2]) = K$.

Solution. Here

$$\Delta = -16 \cdot 4(-\pi^2)^3 = 64\pi^6, \qquad v(\Delta) = 6 < 12,$$

since p is odd. So the equation is minimal (Proposition 2.5) and E has bad reduction. Over $K(\sqrt{\pi})$, E is isomorphic to $E_1\colon y_1^2 = x_1^3 - x_1$, as shown in (5)(b), and E_1 has $\Delta = 64$, a unit: E has potential good reduction. And $x^3 - \pi^2 x = x(x-\pi)(x+\pi)$, so the points of order two $(0,0)$, $(\pm\pi, 0)$ are rational, and $K(E[2]) = K$.

Remark. Compare: $j(E) = 1728 \in \mathcal{O}_K$, and [Sil09, Proposition VII.5.5].

(b) Show that I_v/J has order 2, and that its nontrivial element acts on every $T_\ell(E)$ as -1.

SOLUTION. Let $E_1 : y_1^2 = x_1^3 - x_1$, with $\Delta = 64$, a unit: it has good reduction over every finite extension of K. Fix a square root $\sqrt{\pi} \in \overline{K}$. The map $\phi : E_1 \to E$, $(x_1, y_1) \mapsto (\pi x_1, \pi\sqrt{\pi}\, y_1)$, is an isomorphism, since

$$(\pi\sqrt{\pi}\, y_1)^2 = \pi^3(x_1^3 - x_1) = (\pi x_1)^3 - \pi^2(\pi x_1).$$

Let $\sigma \in \mathrm{Gal}_K$ and $\chi(\sigma) \coloneqq \sigma(\sqrt{\pi})/\sqrt{\pi} \in \{\pm 1\}$. Then $\sigma(\phi(P)) = \phi(\chi(\sigma)\,\sigma(P))$ for $P \in E_1(\overline{K})$, because $(x_1, y_1) \mapsto (x_1, -y_1)$ is $[-1]$. If $\sigma \in I_v$, then σ fixes $E_1[m](\overline{K})$ for every m with $p \nmid m$ (Proposition 2.8), so σ acts on $E[m](\overline{K}) = \phi(E_1[m](\overline{K}))$, and on every $T_\ell(E)$, as multiplication by $\chi(\sigma)$. Since $t^2 - \pi$ is Eisenstein, $K(\sqrt{\pi})/K$ is ramified, so χ is nontrivial on I_v. Hence $J = I_v \cap \ker \chi$ has index 2, and the other coset acts as -1.

(c) So (3) and (4) fail for $m = 2$. Where did your proofs use $m \geqslant 3$?

SOLUTION. **In (3), to find a prime power $\ell^r \geqslant 3$ dividing m.** For $m = 2$, $K(E[2]) = K$, so $I_v \cap \mathrm{Gal}_{K(E[2])} = I_v \neq J$, and (3) fails. And $K(E[2])/K$ is unramified although E has bad reduction, so (4) fails. The proof of (3) needed a prime power $\ell^r \geqslant 3$ dividing m, which does not exist for $m = 2$: the element $-I \in \mathrm{GL}_2(\mathbb{Z}_2)$ has order 2 and is congruent to I modulo 2, but not modulo 4.

Remark. This is why Silverman's hint works modulo 4 when $\ell = 2$.

(6) Suppose that $p \geqslant 5$. Show that $K(E[m])/K$ is tamely ramified for every $m \geqslant 1$ with $p \nmid m$.

SOLUTION. Since $p \geqslant 5$, (3) applies with $m = 3$: I_v/J is isomorphic to the inertia group of $K(E[3])/K$, a subgroup of $\mathrm{Gal}(K(E[3])/K)$, which acts faithfully on $E[3](\overline{K}) \cong (\mathbb{Z}/3\mathbb{Z})^2$ (Corollary 2.6). So I_v/J embeds in $\mathrm{GL}_2(\mathbb{F}_3)$, of order $48 = 2^4 \cdot 3$, and its order is prime to p. For every m with $p \nmid m$, J acts trivially on $E[m](\overline{K})$, so $J \subseteq I_v \cap \mathrm{Gal}_{K(E[m])}$, and the inertia group of $K(E[m])/K$, isomorphic to $I_v/(I_v \cap \mathrm{Gal}_{K(E[m])})$, is a quotient of I_v/J. So its order is prime to p too.

(7) `Magma` computes $\#(I_v/J)$ for elliptic curves over $\mathbb{Q}_7$:

```
p := 7;
Inertia := func< A, B |
  #InertiaGroup(GaloisRepresentation(EllipticCurve([A, B]), p)) >;
Inertia(-1, 0);                    // 1
Inertia(-p^2, 0);                  // 2
Inertia(0, p);                     // 6
```

Check the first two values against your answers, and the third against (6).

SOLUTION. **They agree.** The first curve has $\Delta = 64$, a unit at 7: good reduction, so $I_v = J$ by (4) and (3). The second is the curve of (5) with $\pi = 7$, and $\#(I_v/J) = 2$. The third, $y^2 = x^3 + 7$, has $\#(I_v/J) = 6$, which divides 48. It acquires good reduction over $L = \mathbb{Q}_7(\varpi)$ with $\varpi^6 = 7$: the substitution $x = \varpi^2 x_1$, $y = \varpi^3 y_1$ gives $y_1^2 = x_1^3 + 1$. As $\mu_6(\overline{\mathbb{Q}}_7) \subseteq \mathbb{Q}_7$, $L/\mathbb{Q}_7$ is Galois, and it is totally ramified of degree 6. Its inertia group $I_v \cap \mathrm{Gal}_L$ lies in J, so I_v/J is a quotient of the inertia group of $L/\mathbb{Q}_7$, of order 6. `Magma` says that it is all of it.

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